

Clue Selection From a Learned Listener

Codenames spymaster, exact expected reward under a learned choice model

1 Setup

As before, let A be the set of our team's unrevealed board words and B the set of all other unrevealed board words, with i indexing A and w indexing B . Selecting a word of A is worth 1, and selecting $w \in B$ costs $c_w > 0$. We choose a clue word and a number k , with $1 \leq k \leq K$, to maximize expected reward.

In place of a noise model on similarities, suppose we have a model that, for a fixed clue word, assigns each board word x a real score s_x , and predicts that a guesser choosing from a set S of unrevealed words picks $x \in S$ with probability

$$\Pr(\text{pick } x \mid S) = \frac{\lambda_x}{\sum_{y \in S} \lambda_y}, \quad \lambda_x = e^{s_x}. \quad (1)$$

The guesser selects words one at a time under (1), removing each selection from S , and stops after k selections or upon selecting a word of B , whichever comes first.

Two things are worth making explicit about (1). First, the scores do not depend on S : one score per word is computed once, and removing a word changes only the normalizing sum. This is Luce's choice axiom, and the model that supplies the scores is fitted under exactly this assumption, so using it here is a matter of consistency rather than of convenience. Second, (1) applied repeatedly defines a distribution over orderings known as the Plackett–Luce model, and the expected reward we want is a sum over that distribution: each first selection opens a subtree of possible second selections with renormalized probabilities, and so on. Enumerating that tree costs $O(|A \cup B|^K)$. Section 2 removes the tree entirely.

2 The exponential race

Lemma 1. *Let $T_x \sim \text{Exp}(\lambda_x)$ be independent, one per word. Then the ordering of the words by increasing T_x has exactly the distribution generated by repeated selection under (1).*

Proof. For independent exponentials, $\Pr(\arg \min_x T_x = x_1) = \lambda_{x_1} / \sum_y \lambda_y$, which is (1) for the first selection. By the memorylessness of the exponential, conditional on $T_{x_1} = t$ the remaining variables $T_x - t$ are again independent exponentials with the same rates λ_x , so the same statement applies to the second selection with x_1 removed, and inductively to the rest. Multiplying the successive factors,

$$\Pr(T_{x_1} < T_{x_2} < \dots < T_{x_n}) = \prod_{j=1}^n \frac{\lambda_{x_j}}{\sum_{l \geq j} \lambda_{x_l}},$$

which is the probability the repeated-selection process produces that ordering. $\square$

Lemma 1 is an identity, not an approximation: the race reproduces the selection tree at every node. What it buys is that the quantities we need are now functions of minima of independent exponentials, which are elementary.

Following the same shape as the noise model, let

$$T = \min_{w \in B} T_w, \quad W = \arg \min_{w \in B} T_w, \quad N = \#\{i \in A : T_i < T\}.$$

So W is the word of B the guesser would reach first, T is when, and N is how many of our words they would take before it. The guesser reveals $\min(k, N)$ words of A , and selects a word of B — necessarily W — if and only if $N < k$. The objective is to maximize $G(k) - P(k)$ for

$$G(k) = \mathbb{E}[\min(k, N)], \quad P(k) = \mathbb{E}[c_W \mathbf{1}\{N < k\}].$$

3 Derivation

Write $\Lambda = \sum_{w \in B} \lambda_w$.

The distribution of T . A minimum of independent exponentials is exponential with the summed rate, so $T \sim \text{Exp}(\Lambda)$ and

$$F_T(t) = 1 - e^{-\Lambda t}, \quad dF_T(t) = \Lambda e^{-\Lambda t} dt. \quad (2)$$

The identity of W . For independent exponentials the minimum and the argument of the minimum are independent, and

$$\Pr(W = w) = \frac{\lambda_w}{\Lambda} \quad \text{for every } w. \quad (3)$$

This is where the learned model is cheaper than the noise model. There, the identity of the first non-team word depended on t and required a hazard ratio at every quadrature point; here it does not depend on t at all, so

$$\bar{c} = \mathbb{E}[c_W] = \frac{1}{\Lambda} \sum_{w \in B} c_w \lambda_w$$

is a single constant, computed once per clue in $O(|B|)$.

The distribution of N given T . T is a function of $(T_w)_{w \in B}$ only, which is independent of $(T_i)_{i \in A}$. Conditioning on $T = t$ therefore leaves each T_i exponential with rate λ_i , and each $i \in A$ independently satisfies $T_i < t$ with probability

$$p_i(t) = 1 - e^{-\lambda_i t}. \quad (4)$$

So given $T = t$, N is a sum of $|A|$ independent Bernoulli variables with success probabilities $p_i(t)$ — a Poisson binomial, as in the noise model.

Assembling. Using $\min(k, N) = \sum_{j=1}^k \mathbf{1}\{N \geq j\}$ and conditioning on T ,

$$G(k) = \sum_{j=1}^k \int_0^\infty \Pr(N \geq j \mid T = t) dF_T(t). \quad (5)$$

For $P(k)$, note that given $T = t$ the count N depends only on $(T_i)_{i \in A}$ while W depends only on $(T_w)_{w \in B}$, so c_W and $\mathbf{1}\{N < k\}$ are conditionally independent given T . Combined with (3), which makes $\mathbb{E}[c_W \mid T = t]$ constant in t , the expectation factors completely:

$$P(k) = \bar{c} \int_0^\infty \Pr(N < k \mid T = t) dF_T(t). \quad (6)$$

4 Numerical evaluation

Both (5) and (6) have the form $\int_0^\infty \psi(t) dF_T(t)$ for a bounded ψ . Unlike the noise model, no truncation is needed: substituting $u = e^{-\Lambda t}$ gives $du = -\Lambda e^{-\Lambda t} dt$, so $dF_T(t) = -du$, and as t runs over $(0, \infty)$, u runs over $(0, 1)$. Hence

$$\int_0^\infty \psi(t) dF_T(t) = \int_0^1 \psi(t(u)) du, \quad t(u) = -\frac{\ln u}{\Lambda}, \quad (7)$$

an integral of a bounded function over the unit interval, which a midpoint rule on a uniform grid handles without any choice of bounds. The whole tail of T is carried exactly by the substitution rather than being cut off and argued to be negligible.

The integrand is bounded but not smooth at $u = 0$: by (8) it carries $u^{\lambda_i/\Lambda}$, whose derivative is unbounded there whenever that ratio is below 1. The midpoint rule therefore converges at roughly first order rather than second, and the remedy is simply more cells, since each one costs a power and nothing else. In practice far fewer are needed than the error alone suggests, because the search consumes the ordering of clues rather than their values, and the ordering stabilizes well before the values do.

The success probabilities take a convenient form in u : from (4),

$$p_i(u) = 1 - u^{\lambda_i/\Lambda}, \quad (8)$$

so each quadrature point costs one power per word of A and no special functions anywhere in the computation.

The conditional probabilities $\Pr(N \geq j \mid T = t)$ follow from the standard recursion for a sum of independent Bernoulli variables. Letting $q_r^{(i)}$ be $\Pr(N = r)$ after the first i words of A ,

$$q_0^{(0)} = 1, \quad q_r^{(i)} = q_r^{(i-1)}(1 - p_i) + q_{r-1}^{(i-1)} p_i,$$

and $\Pr(N \geq j) = 1 - \sum_{r < j} q_r$. Counts above K are never used, so the recursion is truncated there and costs $O(|A|K)$ per quadrature point.

With M cells the total is $O(M|A|K + |B|)$ per clue, against $O(M(|A|K + |B|))$ for the noise model: the $|B|$ term leaves the integral because of (3). Every value of k from 1

to K is obtained from one pass, since (5) accumulates over j and (6) needs the same truncated distribution of N .

5 What is exact and what is assumed

It is worth separating the two, because the derivation above is exact in a way that can disguise where the modelling risk actually sits.

Exact, given (1): Lemma 1, so the race is the selection tree rather than a stand-in for it; the distribution (2) of T ; the independence and law (3) of W ; the conditional Poisson binomial (4); the factorization of $P(k)$; the substitution (7); and the Bernoulli recursion. The cap $\min(k, N)$ is likewise exact and not a truncation, since $N \geq k$ is precisely the event that the guesser exhausts its guesses on our words and never reaches W . The sole numerical error is the quadrature in (7), over a bounded integrand on a bounded interval.

Assumed, and not established here:

Scores do not depend on the remaining set. This is the content of (1) and the one substantive modelling commitment. It is not a cost of the race construction — a direct enumeration of the selection tree assumes the same thing the moment it renormalizes the scores over the survivors rather than recomputing them. If the guesser's scores genuinely change as the board empties, both routes are wrong together, and Plackett–Luce is the wrong family rather than the integral being the wrong evaluation of it.

The guesser spends every guess. The process stops only on reaching k selections or a word of B . A guesser who declines a guess it is unsure of faces a different expected reward, and nothing in the listener model expresses that choice.

The clue number is the guess budget. k enters only as the cap. Where the rules grant a bonus guess, K should be set to the true budget rather than to the announced number.